Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Given that $y = A\sin\left[\left(\frac{2\pi}{\lambda}(ct - x)\right)\right]$, where y and x are measured in metres. Which of the following statements is true?

A. The unit of λ is same as that of x and A

B. The unit of (ct -x) is same as that of $\frac{2\pi}{\lambda}$

C. The unit of c is same as that of $\frac{2\pi}{\lambda}$

D. The unit of λ is same as that of x but not of A

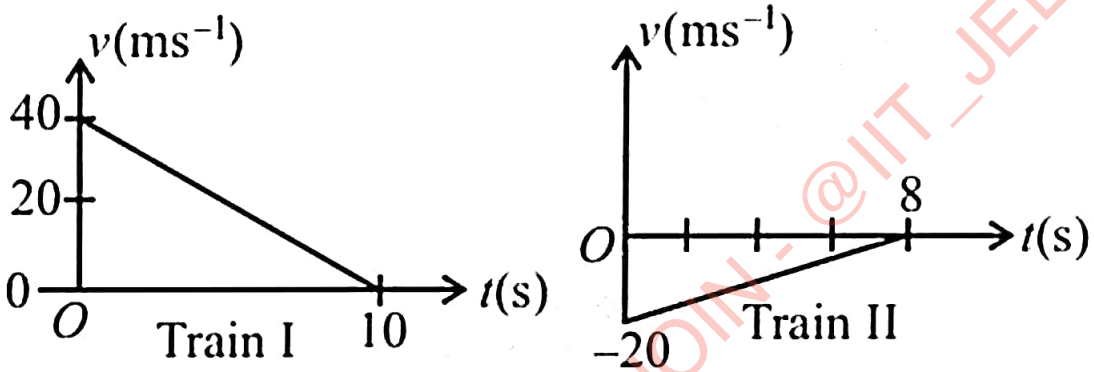
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Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two trains, which are moving along different tracks in opposite directions are put on the same track by mistake. On noticing the mistake, when the trains are 300 m apart the drivers start slowing down the trains. The graphs given below show decrease in their velocities as function of time. The separation between the trains when both have stopped is



A. $120\ m$

B. $20\ m$

C. $60\ m$

D. $280\ m$

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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A body is projected vertically upwards. The times corresponding to height while ascending and while descending are t_1 and t_2 respectively. Then the velocity of projection is (g is acceleration due to gravity)

A. $g\sqrt{t_1 t_2}$

B. $\frac{gt_1 t_2}{t_1+t_2}$

C. $\frac{g\sqrt{t_1 t_2}}{2}$

D. $\frac{g(t_1+t_2)}{2}$

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Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion: Although both, a stone and a feather, are dropped from the same elevation, they do not reach the ground at the same time.
Reason: Acceleration due to gravity, depending upon the mass of the object, varies from object to object.
Choose the correct option:

A. If both assertion and reason are true and reason is the correct explanation of assertion.

B. If both assertion and reason are true but reason is not the correct explanation of assertion.

C. If assertion is true but reason is false.

D. If both assertion and reason are false.

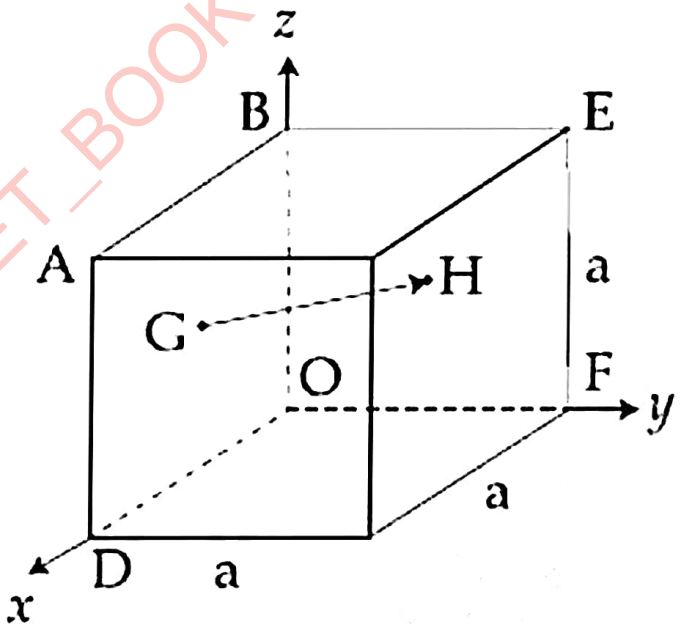
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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the cube of side 'a' shown in the figure, the vector from the central point of the face $ABOD$ to the central point of the face $BEFO$ will be:



A. $\frac{1}{2}\, a\, \left(\hat{k} - \hat{i}\right)$

B. $\frac{1}{2}\, a \left(\hat{i} - \hat{k}\right)$

C. $\frac{1}{2}\, a \left(\hat{j} - \hat{i}\right)$

D. $\frac{1}{2}\, a \left(\hat{j} - \hat{k}\right)$

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Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the energy, $E = G^p h^q c^r$, where G is the universal gravitational constant, h is the Planck's constant and c is the velocity of light, then the values of p, q and r are, respectively

A. $-\frac{1}{2}, \frac{1}{2}$ and $\frac{5}{2}$

B. $\frac{1}{2}, -\frac{1}{2}$ and $-\frac{5}{2}$

C. $-\frac{1}{2}, \frac{1}{2}$ and $\frac{3}{2}$

D. $\frac{1}{2}, -\frac{1}{2}$ and $-\frac{3}{2}$

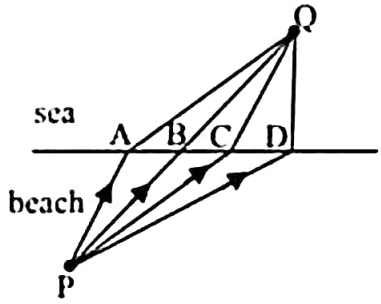
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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A girl standing at point P on a beach wishes to reach a point Q in the sea as quickly as possible. She can run at $6\ \mathrm{km\ h^{-1}}$ on the beach and swim at $4\ \mathrm{km\ h^{-1}}$ in the sea. She should take the path



A. PAQ

B. PBQ

C. PCQ

D. PDQ

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Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A person can throw a stone to a maximum distance of 100 m. The greatest height to which he can throw the stone is:-

A. 100 m

B. 75 m

C. 50 m

D. 25 m

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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Three balls each of mass1 kg are attached with three strings each of length 1 m as shown in figure. They are rotated in a horizontal circle with angular velocity $\omega = 4$ rad s^{-1} about point O. Match the following.

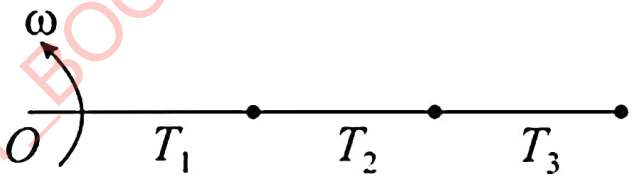


	Table 1		Table 2
(A)	T_1	(P)	Maximum
(B)	T_2	(Q)	Minimum
(C)	T_3	(R)	80 N
		(S)	45 N
		(T)	48 N
		(U)	90 N

A. (A) - P, (B) - R, (C) - Q, T

B. (A) - T, (B) - R, (C) - Q, P

C. (A) - S, (B) - P, (C) - Q, T

D. (A) - Q, (B) - R, (C) - S, T

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Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A body, starting from rest, is moved along a straight line by a machine delivering constant power. The distance moved by the body in time t is proportional to (given: power = force $\times$ velocity)

A. $t^{1/2}$

B. $t^{3/4}$

C. $t^{3/2}$

D. $t^{1/4}$

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Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A ball falls vertically downward and bounces off a horizontal floor. The speed of the ball just before reaching the floor (u_1) is equal to the speed just after having contact with the floor (u_2); $u_1 = u_2$. The corresponding magnitudes of accelerations are denoted respectively by a_1 and a_2. The air resistance during motion is proportional to speed and is not negligible. If g is acceleration due to gravity, then-

A. $a_1 < a_2$

B. $a_1 > a_2$

C. $a_1 = a_2 \neq g$

D. $a_1 = a_2 = g$

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Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A particle is in uniform circular motion, related to one complete revolution of the particle, which among the statements is incorrect?

A. Average acceleration of the particle is zero.

B. Displacement of the particle is zero.

C. Average speed of the particle is zero.

D. Average velocity of the particle is zero.

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Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The Quantum Hall Resistance R_H is a fundamental constant with dimensions of resistance. If h is Planck's constant and e the electron charge, then the dimension of R_H is the same as -

A. e^2/h

B. h/e^2

C. h^2/e

D. e/h^2

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Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two skaters P and Q are skating towards each other. Skater P throws a ball towards Q every 5 s such that it always leaves her hand with speed $2\ \mathrm{ms}^{-1}$ with respect to the ground. Consider two cases:
(i) P runs with speed $1\ \mathrm{ms}^{-1}$ towards Q while Q remains stationary.
(ii) Q runs with speed $1\ \mathrm{ms}^{-1}$ towards P while P remains stationary.
Note that irrespective of speed of P, ball always leaves P s hand with speed $2\ \mathrm{ms}^{-1}$ with respect to the ground. Ignore gravity. Balls will be received by Q

A. One every $2s$ in case (I) and one every 4 s in case(II)

B. One every 3.35 in case (I) and one every 2.55 in case (II)

C. One every 2.55 in case (I) and one every 2.5 s in case (II)

D. One every 2.55 in case (I) and one every 3.35 in case (II)

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Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If C be the capacitance and V be the electric potential, then the dimensional formula of CV^2 is

A. $\left[\mathrm{ML^2\ T^{-2}\ A^0}\right]$

B. $\left[\mathrm{MLT^{-2}\ A^{-1}}\right]$

C. $\left[\mathrm{M^0LT^2\ A^0}\right]$

D. $\left[\mathrm{ML^3TA}\right]$

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Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two guns P and Q can fire bullets at speeds $2\ \mathrm{km/s}$ and $4\ \mathrm{km/s}$, respectively. From a point on a horizontal ground, they are fired in all possible directions. The ratio of maximum areas covered by the bullets fired by the two guns, on the ground is

A. $1:2$

B. $1:4$

C. $1:8$

D. $1:16$

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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A stone falls freely such that the distance covered by it in the last second of its motion is equal to the distance covered by it in the first 5 s. It is in air for..... s.

A. 12

B. 13

C. 25

D. 26

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Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The maximum range of a gun on horizontal plane is $16\ \text{km}$. If $g = 10\ \text{ms}^{-2}$, then muzzle velocity of a shell is

A. $160\ \text{ms}^{-1}$

B. $200\sqrt{2}\ \text{ms}^{-1}$

C. $400\ \text{ms}^{-1}$

D. $800\ \text{ms}^{-1}$

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Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Given the numbers : 161 cm, 0.161 cm, 0.0161 cm. The number of significant figures for the three numbers are

A. 3,4 and 5 respectively

B. 3,3 and 4 respectively

C. 3,3 and 3 respectively

D. 3,4 and 4 respectively

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Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A girl drops an apple from the window of a train which is moving on a straight track with speed increasing with a constan rate. The trajectory of the falling apple as seen by the girl is

A. parabolic and in the direction of the moving train

B. parabolic and opposite to the direction of the moving train

C. an inclined straight line pointing in the direction of the moving train

D. an inclined straight line pointing opposite to the direction of the moving train

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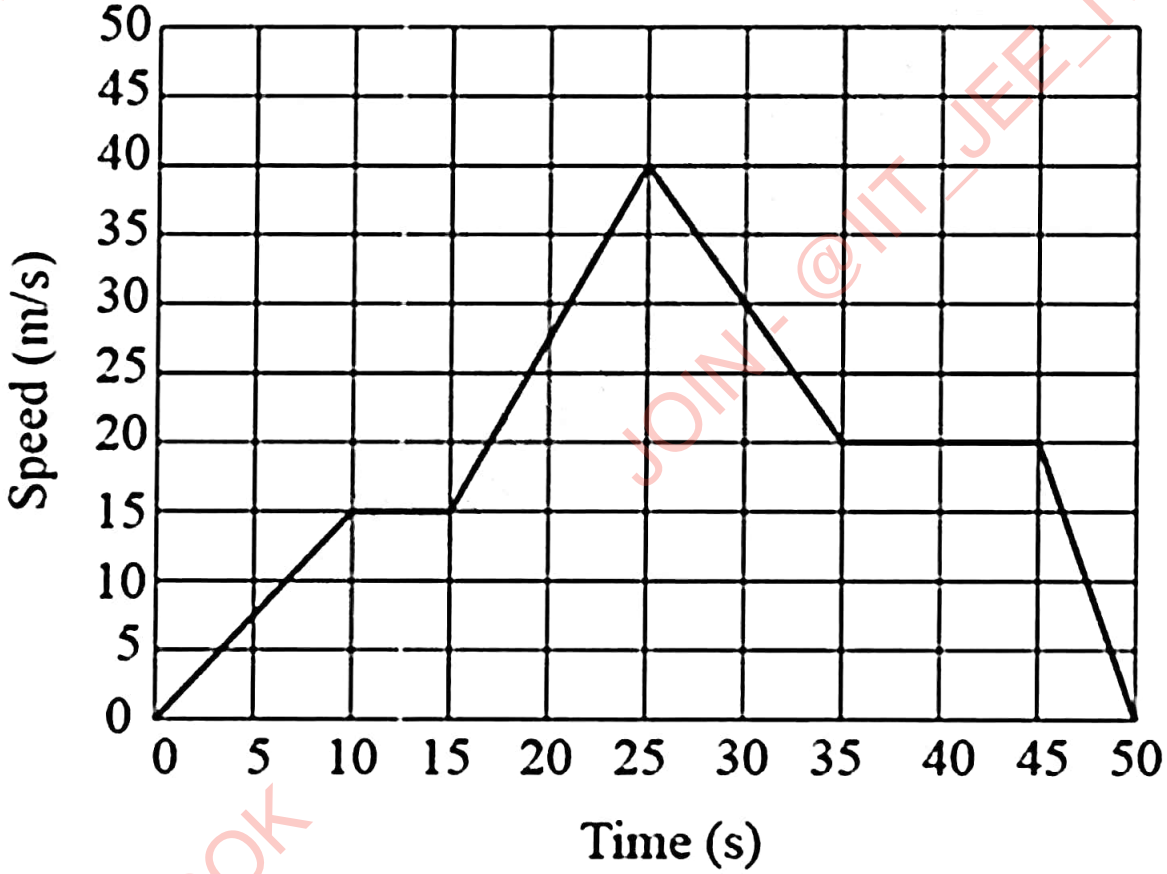
Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

The following figure gives the movement of an object.



Find total distance travelled by object.

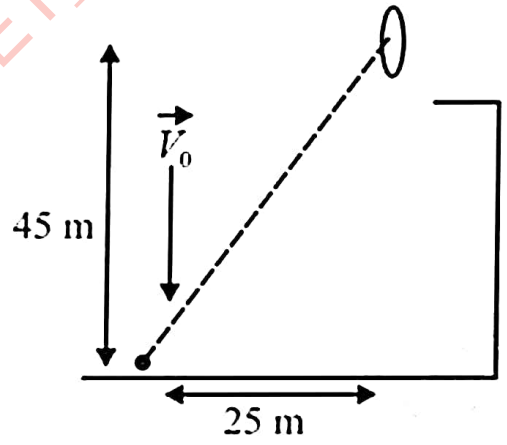
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Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

In a circus, a performer throws an apple towards a hoop held at 45 m height by another performer standing on a high platform (see figure below). The thrower aim s for the hoop and throws the apple with a speed of $24\ \text{m s}^{-1}$. At the exact moment that the thrower released the apple, the other performer drops the hoop. The hoop falls straight down. At what height above the ground does the apple go through the hoop?



(roundoff answer to nearest integer)

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

Two balls are projected with the same velocity but with different angles with the horizontal. Their ranges are equal. If the angle of projection of one is 30° and its maximum height is h, then the maximum height of other will be _____ h.

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

A physical quantity $X = \frac{A^2 B}{C^{1/3}\sqrt{D}}$ is calculated by using measured quantities A, B, C and D. If the errors in the measurement of A, B, C and D are $1\%, 2\%, 3\%$ and 4% respectively, find the percentage error in the measurement of X.

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

A motorboat covers a given distance in 6 h moving downstream on a river. It covers the same distance in 10 h moving upstream. The time it takes to cover the same distance in still water is K hours, find 2K.

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Question No. 6

Numerical Type

The answer has to be filled into the input box provided below.

The height y and the distance x along the horizontal plane of a projectile on a certain planet (with no surrounding atmosphere) are given by $y = 8t - 5t^2$ m and $x = 6t$ m, where t is in seconds. The velocity with which the projectile is projected is

Question No. 7

Numerical Type

The answer has to be filled into the input box provided below.

If unit of length, mass and time each be doubled, the new unit of work done is _____ times the old unit of work

Question No. 8

Numerical Type

The answer has to be filled into the input box provided below.

A mouse jumps off from the 15th floor of high-rise building and lands 12 m from the building. Assume that each floor is of 3 m height. The horizontal speed with which the mouse jumps is v km/hr, find value of 20v.

Question No. 9

Numerical Type

The answer has to be filled into the input box provided below.

A person goes from point P to point Q covergin $1/3$ of the distance with speed 10 km/hr, the next $1/3$ of the distance at 20 km/hr and the last $1/3$ of the distance at 60 km/hr. The average speed of the person is (Mark answer in km/hr)

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Question No. 9

Numerical Type

The answer has to be filled into the input box provided below.

A person goes from point P to point Q covergin $1/3$ of the distance with speed $10\ \mathrm{km/hr}$, the next $1/3$ of the distance at $20\ \mathrm{km/hr}$ and the last $1/3$ of the distance at $60\ \mathrm{km/hr}$. The average speed of the person is (Mark answer in km/hr)

Question No. 10

Numerical Type

The answer has to be filled into the input box provided below.

The displacement x (in m) of a particle of mass m (in kg) moving in one dimension under the action of a force, is related to time t (in sec) by $t = \sqrt{x} + 3$. The displacement of the particle when its velocity is zero, will be

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Chemistry Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

One among the following is an incorrect statement-

A. Molality of a solution is dependent on the temperature

B. Molarity of a solution is dependent on the temperature

C. Normality of $0.5\ M$ aqueous solution of $H_2C_2O_4 \cdot 2H_2O$ is $1\ N$

D. Molality of a solution relates moles of solute and mass of solvent

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Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An electron is moving in Bohr's fourth orbit. Its de-Broglie wave length is λ. What is the circumference of the fourth orbit?

A. $\frac{2}{\lambda}$

B. 2λ

C. 4λ

D. $\frac{4}{\lambda}$

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The reaction,
$K_2Cr_2O_7 + mFeSO_4 + nH_2SO_4 \rightarrow Cr_2(SO_4)_3 + pFe_2(SO_4)_3 + H$
when balanced, m, n, p, and q are, respectively

A. $6, 14, 3, 14$

B. $6, 7, 3, 7$

C. $3, 7, 2, 7$

D. $4, 14, 2, 14$

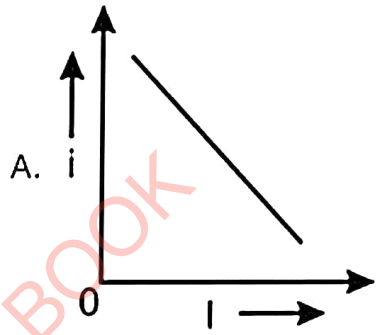
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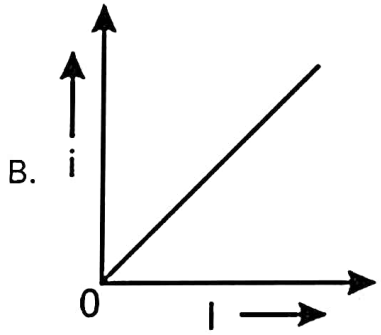
Question No. 4

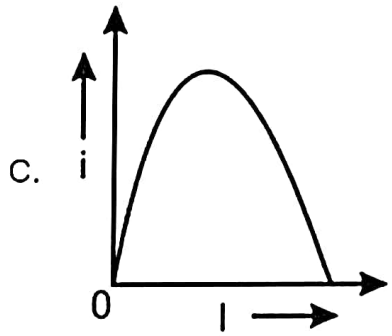
Only One Option Correct Type

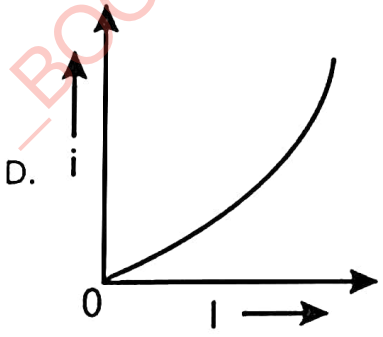
Each question has multiple options out of which ONLY ONE is correct.

Which of the following graphs represents the variation of photoelectric current (i) with intensity (I) of the incident light?

A. 

B. 

C. 

D. 

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Natural abundances of ^{12}C and ^{13}C isotopes of carbon are 99% and 1%, respectively. Assuming they only contributes to the mol. wt. of C_2F_4, the percentage of C_2F_4 having a molecular mass of 101 is

A. 1.98

B. 98

C. 0.198

D. 99

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Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The correct statements among the following.

i. $E_{2s}(\mathrm{H}) > E_{2s}(\mathrm{Li}) > E_{2s}(\mathrm{Na}) > E_{2\delta}(\mathrm{K})$

ii. The maximum number of electrons in the shell with principal quantum number n is equal to $2n^2$

iii. Extra stability of half-filled subshell is due to smaller exchange energy.

iv. Only two electrons, irrespective of their spin, may exist in the same orbital are.

A. i and ii

B. ii and iii

C. iii and iv

D. i and iv

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Among the following, the correct statement about cathode ray discharge tube is

A. the electrical discharge can only be observed at high pressure and at low voltage.

B. in the absence of external electrical or magnetic field, cathode rays travel in straight lines.

C. the characteristics of cathode rays depend upon the material of electrodes.

D. the characteristics of cathode rays depend upon the gas present in the cathode ray tube.

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Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

1.25 g of a metal (M) reacts with oxygen completely to produce 1.68 g of metal oxide.
The empirical formula of the metal oxide is [molar mass of M and O are $69.7 \mathrm{gmol}^{-1}$ and $16.0 \mathrm{gmol}^{-1}$, respectively]

A. $\mathrm{M_2O}$

B. $\mathrm{M_2O_3}$

C. $\mathrm{MO_2}$

D. $\mathrm{M_3O_4}$

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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

When 22.4 L of C_4H_8 at STP is burnt completely, 89.6 L of CO_2 gas at STP and 72 g of water are produced. The volume of the oxygen gas at STP consumed in the reaction is closest to :

A. 89.6 L

B. 112 L

C. 134.4 L

D. 22.4 L

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Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

What is the lowest energy of the spectral line emitted by the hydrogen atom in the Lyman series?

(h = Planck's constant, c = velocity of light, R = Rydberg's constant).

A. $\frac{5hcR}{36}$

B. $\frac{4hcR}{3}$

C. $\frac{3hcR}{4}$

D. $\frac{7hcR}{144}$

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Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Alcohol level in blood can be determined by a redox titration with dichromate solution in acidic medium according to the balanced equation

$$C_2H_5OH(aq) + 2Cr_2O_7^{2-}(aq) + 16H^+ \longrightarrow 2CO_2(g) + 4Cr^{3+}$$

What is blood alcohol level in mass percent if 8.0 mL of 0.05 M $Cr_2O_7^{2-}$ solution is required for titration of a 10.0 g sample of blood?

A. 0.092%

B. 0.01%

C. 1.4%

D. 1.2%

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Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion : Between any two given energy levels, the number of absorption transitions is always less than the number of emission transitions.
Reason: Absorption transitions start from the lowest energy level only and may end at any higher energy level. But emission transitions may start from any higher energy level and end at an energy level below it.

A. Both Assertion and Reason are correct and Reason is the correct explanation for Assertion

B. Assertion is correct but Reason is incorrect

C. Assertion is incorrect but Reason is correct

D. Both Assertion and Reason are correct bat Reason is not the correct explanation for Assertion

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Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match the following.

List-I (Type of reaction)		List-II (Unbalanced equation)
A. Combination reaction	(i)	$AgNO_3 + CaCl_2 \longrightarrow AgCl + Ca(NO_3)_2$
B. Decomposition reaction	(ii)	$Cl_2 \longrightarrow Cl^- + ClO_3^-$
C. Disproportionation reaction	(iii)	$Mg + N_2 \longrightarrow Mg_3N_2$
D. Displacement reaction	(iv)	$KClO_3 \xrightarrow{\Delta} KCl + O_2$

A. A-(iii); B-(ii); C-(iv); D-(i)

B. A-(i); B-(iv); C-(ii); D-(iii)

C. A-(i); B-(ii); C-(iii); D-(iv)

D. A-(iii); B-(iv); C-(ii); D-(i)

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Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The energy associated with Bohr's orbit in the hydrogen atom is given by the expression, $E_n = -\frac{13.6}{n^2}$ eV. The energy in eV associated with the orbit having a radius $9r_1$ is (r_1 is the radius of the first orbit)

A. -13 6

B. -6.8

C. -1.51

D. -1.36

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An aqueous solution of alcohol contains 18 g of water and 414 g of ethyl alcohol. The mole fraction of water is

A. 0.4

B. 0.7

C. 0.9

D. 0.1

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Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The empirical formula and molecular mass of a compound are CH_2O and 180 g respectively. The molecular formula of the compound will be

A. $C_9H_{18}O_9$

B. CH_2O

C. $C_6H_{12}O_6$

D. $C_2H_4O_2$

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A certain metal has a work function of $\Phi = 2\text{eV}$. It is irradiated first with 1 W of 400nm light and later with 1 W of 800nm light. Among the following, the correct statement is:
[Given:Planck constant $(\text{h}) = 6.626 \times 10^{-34}\ \text{m}^2\text{kgs}^{-1}$ Speed of light $\left(\text{c}\right) = 3 \times 10^8\ \text{ms}^{-1}$]

A. Both colors of light give rise to same number of photoelectrons.

B. 400nm light gives rise to less energetic photoelectrons than 800nm light.

C. 400nm light leads to more photoelectrons.

D. 800nm light leads to more photoelectrons.

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Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The oxidation number of nitrogen atoms in NH_4NO_3 are

A. $+5, +5$

B. $-3, +5$

C. $+3, -5$

D. $-3, -3$

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Calculate the maximum number of electrons of an atom which will have the following set of quantum numbers?

$n = 4,\ \ell = 0,\ 1,\ 2,\ m = 0,\ \pm 1$ and $s = \pm 1/2$.

A. 12

B. 7

C. 18

D. 14

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Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Find the values of x, y, z in the given balanced equation
$H_2SO_4 + xHI \rightarrow H_2S + yI_2 + zH_2O$

A. $x = 3, y = 5, z = 2$

B. $x = 4, y = 8, z = 5$

C. $x = 8, y = 4, z = 4$

D. $x = 5, y = 3, z = 4$

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Chemistry Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

3.92 g of ferrous ammonium sulphate react completely with 50ml of $\frac{N}{10}$ $KMnO_4$ solution. The percentage purity of the sample is

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

Upon fully dissolving 2.0 g of a metal in sulfuric acid, 6.8 g of the metal sulfale is formed. The equivalent weight of the metal is —

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

The maximum number of electrons that can have principal quantum number, $n = 3$, and spin quantum number, $m_s = -\frac{1}{2}$, is

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

The uncertainties in the velocities of two particles A and B are 0.05 and 0.02 m/s^{-1} respectively. The mass of B is five times to that of mass A. What is the ratio of uncertainties $\left(\frac{\Delta x_A}{\Delta x_B}\right)$ in their positions?

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

The normality of 26% (wt/vol) solution of ammonia (density $= 0.855$) is approximately: (roundoff answer to nearest integer.)

Question No. 6

Numerical Type

The answer has to be filled into the input box provided below.

The ratio of lowest energy in terms of wave numbers of Balmer and Lyman series of lines of atomic spectrum of hydrogen is 10:K, find K.

Question No. 7

Numerical Type

The answer has to be filled into the input box provided below.

4 mol of a mixture of Mohr's salt and $Fe_2(SO_4)_3$ requires 500ml of 1M, $K_2Cr_2O_7$ for complete oxidation in acidic medium. The mole % of the Mohr's salt in the mixture is

Question No. 8

Numerical Type

The answer has to be filled into the input box provided below.

A mixture of $CaCl_2$ and NaCl weighing 4.44 g is treated with sodium carbonate solution to precipitate all the calcium ions as calcium carbonate. The calcium carbonate so obtained is heated strongly to get 0.56 g of CaO. The percentage of NaCl in the mixture is (atomic mass of Ca = 40)

Question No. 9

Numerical Type

The answer has to be filled into the input box provided below.

How many of the following orbitals have two angular and one radial nodes?

3d; 4d; 5f; 3p

Question No. 10

Numerical Type

The answer has to be filled into the input box provided below.

The vapour density of ozone is

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Mathematics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the complex number z satisfy the equation $(i-z)(1+2i)+(1-iz)(3-4i)=1+7i$, then $z+\overline{z}+z\overline{z}$ is equal to -

A. 0

B. 1

C. -1

D. None of these

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The set of all solution of inequality $\frac{(e^x-1)(2x-3)(x^2+x+2)}{(\sin x-\sqrt{5})(x+1)x} \leq 0$ is

A. $\left[\frac{3}{2}, \infty\right)$

B. $\left(-\infty, -1\right) \cup \left[\frac{3}{2}, \infty\right)$

C. $\left(-1, 0\right) \cup \left[\frac{3}{2}, \infty\right)$

D. $R-\{0,-1\}$

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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $a > 2$, then the roots of the equation $(2 - a)x^2 + 3ax - 1 = 0$ are

A. one positive and one negative

B. both negative

C. both positive

D. both imaginary

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If z is a complex number, then curves $|z| = 1$, $|z - 2| = 1$ and $|z - 1| = 0$ have a common point at

A. $(0, 1)$

B. $(2, 0)$

C. $(1, 0)$

D. $(0, 2)$

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The number of integral values of c for which both the roots of the equation $x^2 - 2cx + c^2 = 1$ are greater than -2 but less than 4 are

A. 0

B. 1

C. 2

D. more than 2

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The minimum value of $f(x) = \frac{x^2-2x+3}{x^2-4x+7}$ is

A. $1 + \frac{1}{\sqrt{3}}$

B. $\frac{3-\sqrt{3}}{3}$

C. $2 - \frac{1}{\sqrt{3}}$

D. $3 - \frac{1}{\sqrt{3}}$

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The complex number z, satisfying the equation $z^3 = \overline{z}$ and $\arg(z+1) = \frac{\pi}{4}$ simultaneously, is (where, $i^2 = -1$)

A. i

B. $1+2i$

C. $2+3i$

D. $3+4i$

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If ω is an imaginary cube root of unity, then the value of $\left(1-\omega+\omega^2\right)\cdot\left(1-\omega^2+\omega^4\right)\cdot\left(1-\omega^4+\omega^8\right)\cdot\ldots$ (2n factors) is

A. 2^{2n}

B. 2^n

C. 1

D. 0

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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $a + b + c > \frac{9c}{4}$ and the equation $ax^2 + 2bx - 5c = 0$ has non-real complex roots, then

A. $a > 0, c > 0$

B. $a > 0, c < 0$

C. $a < 0, c < 0$

D. $a < 0, c > 0$

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If α & β are the roots of the equation $x^2 - p(x+1) - q = 0$, then the value of $\frac{\alpha^2+2\alpha+1}{\alpha^2+2\alpha+q} + \frac{\beta^2+2\beta+1}{\beta^2+2\beta+q}$ is

A. 2

B. 1

C. 0

D. none of these

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Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If z is a complex number such that $z^2 = (\bar{z})^2$, then

A. z is purely real

B. z is purely imaginary

C. either z is purely real or purely imaginary

D. None of these

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If α, β are roots of the equation $x^2 + 5\left(\sqrt{2}\right)x + 10 = 0, \alpha > \beta$ and $P_n = \alpha^n - \beta^n$ for each positive integer n, then the value of $\left(\frac{P_{17}P_{20}+5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19}+5\sqrt{2}P_{18}^2}\right)$ is equal to

A. 2.5

B. 5

C. 1

D. $\sqrt{2}$

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Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If x satisfies the inequality $\log_{25} x^2 + \left(\log_5 x\right)^2 < 2$, then x belongs to

A. $\left(\frac{1}{5}, 5\right)$

B. $\left(\frac{1}{25}, 5\right)$

C. $\left(\frac{1}{5}, 25\right)$

D. $\left(\frac{1}{25}, 25\right)$

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Suppose $\alpha,\ \beta,\ \gamma$ are the roots of $x^3 + x^2 + x + 2 = 0$ then the value of $\left(\frac{\alpha+\beta-2\gamma}{\gamma}\right)\left(\frac{\beta+\gamma-2a}{\alpha}\right)\left(\frac{\gamma+\alpha-2\beta}{\beta}\right)$ is

A. $-\frac{47}{2}$

B. $\frac{47}{2}$

C. -47

D. 47

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Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If z and w are complex numbers satisfying $\bar{z} + i\bar{w} = 0$ and $\arg(zw) = \pi$, then $\arg(w)$ is equal to (where, $\arg(w) \in (-\pi, \pi]$)

A. $\frac{\pi}{4}$

B. $\frac{-\pi}{4}$

C. $\frac{\pi}{2}$

D. $\frac{3\pi}{4}$

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The sum of the solutions of the equation $|\sqrt{x} - 2| + \sqrt{x}(\sqrt{x} - 4) + 2 = 0, (x > 0)$ is equal to

A. 10

B. 9

C. 12

D. 4

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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the real part of $\frac{\bar{z}+2}{\bar{z}-1}$ is $4, z \neq 1$, then the locus of the point representing z in the complex plane is

A. a straight line parallel to x -axis

B. a straight line equally inclined to axes

C. a circle with radius 2

D. a circle with radius $\frac{1}{2}$

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $(3+2\sqrt{2})^{x^2-4}+(3-2\sqrt{2})^{x^2-4}=6$, then $x^4+x^2+5=$

A. -30

B. -35

C. 30

D. 35

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Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $X_n = \left\{ z : \ z = x + iy; \ x, y \in R \ \text{and} \ |z|^2 \leq \frac{1}{n} \right\}$ for all integers $n \geq 1$. Then, $\bigcap_{n=1}^{\infty} X_n$ is

A. a singleton set

B. an empty set

C. an finite set with more than one element

D. none of these

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$\frac{x-1}{3x+4} < \frac{x-3}{3x-2}$ holds, for all x in the interval

A. $\left(\frac{-4}{3} , \frac{2}{3} \right)$

B. $\left(-\infty, \frac{-5}{4} \right)$

C. $(-\infty)$

D. $\left(-\infty, \frac{-5}{4} \right) \cup \left(\frac{3}{4} , -\infty \right)$

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Mathematics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

If $\mathrm{A} = \begin{bmatrix} \frac{-1+i\sqrt{3}}{2i} & \frac{-1-i\sqrt{3}}{2i} \\ \frac{1+i\sqrt{3}}{2i} & \frac{1-i\sqrt{3}}{2i} \end{bmatrix}$, $i = \sqrt{-1}$ and $f(x) = x^2 + 2$. Then $\left| \frac{f(\mathrm{A})}{(2+i\sqrt{3})} \right|$ is equal to

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

If $\frac{\alpha+5i}{2}$ is a root of the equation $2x^2 - 6x + k = 0$, then the value of k is $(\alpha, k, \in R)$

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

If $a > b > 0$ and $a^3 + b^3 + 27ab = 729$, then the quadratic equation $ax^2 + bx - 9 = 0$ has roots α & β, $(\alpha < \beta)$. Find the value of $4\beta - a\alpha$.

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

Let $z = x + yi$, where, x, y are integers and $i = \sqrt{-1}$ the area of the rectangle whose vertices are the roots of the equation $\bar{z}z^3 + z(\bar{z})^3 = 350$ is

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

α, β be the roots of $x^2 - 3x + a = 0$ and γ, δ be the roots of $x^2 - 12x + b = 0$ and numbers $\alpha, \beta, \gamma, \delta$ (in order) form an increasing G.P. then the value of ab is

Question No. 6

Numerical Type

The answer has to be filled into the input box provided below.

Let A, B be two sets of complex numbers as defined below

$A = \{z : Im(z) = |Re(z)|\}$

$B = \left\{z : arg\left(\frac{z-2}{z+2}\right) = \frac{\pi}{2}\right\}$

Find the number of common points in set A and B

Question No. 7

Numerical Type

The answer has to be filled into the input box provided below.

If z_1, z_2 are two complex numbers satisfying $\left|\frac{z_1 - 3z_2}{3 - z_1\bar{z}_2}\right| = 1, |z_1| \neq 3$, then $|z_2|$ is equal to

Question No. 8

Numerical Type

The answer has to be filled into the input box provided below.

The smallest positive integral value of n such that

$\left[\frac{1+\sin\frac{\pi}{8}+i\cos\frac{\pi}{8}}{1+\sin\frac{\pi}{8}-i\cos\frac{\pi}{8}}\right]^n$ is purely imaginary, is equal to

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Question No. 7

Numerical Type

The answer has to be filled into the input box provided below.

If z_1, z_2 are two complex numbers satisfying $\left|\frac{z_1-3z_2}{3-z_1\bar{z}_2}\right| = 1, |z_1| \neq 3$, then $|z_2|$ is equal to

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The smallest positive integral value of n such that

$$\left[\frac{1+\sin\frac{\pi}{8}+i\cos\frac{\pi}{8}}{1+\sin\frac{\pi}{8}-i\cos\frac{\pi}{8}}\right]^n$$

is purely imaginary, is equal to

Question No. 9

Numerical Type

The answer has to be filled into the input box provided below.

The number of real roots of the equation $e^{4x} - e^{3x} - 4e^{2x} - e^x + 1 = 0$ is equal to

Question No. 10

Numerical Type

The answer has to be filled into the input box provided below.

If $x + \log_{10}(1 + 2^x) = x\log_{10}5 + \log_{10}6$, then the value of x is